

WRRRA M 0 10 Address Information Load and Physical Energy and Pressure

Sector exchange gravity and expansion calculated from one energy functional

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Verification inputs and calculation scope

WRRRA studies one nonprivileged universe through minimal computation, a common carrier and phenotype. WRRRA_M specializes the frozen WRRRA Core 1.0 and Minimal Computation Cosmology 2.3.2 to validated inputs and adopted rules for our universe. Version 0.10 supplies positive SI energy responses to the executable 0.9 address ledger and calculates pressure by differentiating that same energy with respect to volume.

Verification inputs are the frozen 0.9-r1 ledger; inherited 0.6-r2 carrier operators, G , H_0 , c , test mass and geometry; and the 0.8 particle selection and charge ledger. The arithmetic partition 5%, 26.8%, 68.2% and the inherited energy calibration 4.93%, 26.5%, 68.57% remain distinct. Address-response coefficients, volume exponents and the product address-carrier state are disclosed constitutive inputs. The defining SI value $c=299792458$ m/s is fixed.

WRRRA specific transformation

The common address state and three sector effects are summed with physical response weights. SI coefficients are calibrated once on a frozen reference and held unchanged when address or carrier states change. The same energy operator generates state updates, while its volume dependence determines pressure. Clustering D density enters the inherited local gravity response, D plus R supplies the twist load, and total density and pressure enter homogeneous expansion.

Outputs

All 32 verification groups pass. The reference returns deceleration -0.52855 , rotation 207.5109051266 km/s and conditional finite-patch deflection 0.5355865106 arcsec. All nine inherited 0.8 state-volume cases reproduce. Outputs include dimensional address energy and pressure, noncommuting sector exchange, admission conversion work and physical responses to changed address rules with calibration held fixed.

Falsification conditions and assessment

The connection must be revised if an admissible state yields negative energy, frozen calibration fails reference reproduction, pressure disagrees with an independent volume derivative, internal exchanges do not cancel, or admission energy changes have no exchange account. Duplicate channel or generation energy and disagreement between text and execution also reject integration. The previously empty 0.9 energy-pressure interface is now executed. Physical measurement outcomes and observation records remain the subsequent 0.13 calculation.

Joint address and carrier inputs

Normalized address weights w_n and sector effects $e_s(n)$ on $n=2\dots 1000000$ are inherited unchanged from 0.9. The sectors are phenotype ϕ , resident nonphenotype D and background response from returned provenance R . Resident Actual is $\phi+D=31.8\%$; the complete downstream ledger including returned response accounts for 100%. The carrier state ρ is a positive unit-trace density matrix on the shared 128-site lattice. The two state spaces are joined by the following product-state construction.

$$\sum_n w_n = 1, e_s(n) \geq 0, \sum_s e_s(n) = 1, \sigma = \sum_n w_n |n\rangle \langle n| \otimes \rho. \quad (1)$$

The address response g_s is a positive constitutive function on the finite address domain. The reference adopts $\lambda_{\phi}=0$, $\lambda_D=0.25$ and $\lambda_R=0.1$. It connects address and sector energy response without identifying n with length or a bit count. Changed address states are summed with the same response and SI coefficients.

$$g_s(n) = 1 + \lambda_s \frac{\log n}{\log N_{\text{address}}}, \mu_s = \sum_n w_n e_s(n) g_s(n). \quad (2)$$

$$A_{\phi} = I, A_D = \frac{K_c}{2}, A_R = \frac{K_b}{2}, L_s = \text{Tr}(\rho A_s). \quad (3)$$

K_c is the inherited positive periodic-lattice Laplacian. K_b is the normalized operator containing the disclosed noncommuting local response. The uniform mixed carrier $\rho = I/N_{\text{carrier}}$ gives all three $L_s=1$. The release inherits the baseline operators and noncommuting test $\epsilon=8$ and retains every positive load, including very small values.

One SI calibration and address energies

The inherited $H_0=67.4$ km/s/Mpc, $G=6.6743 \times 10^{-11}$ m³/kg/s² and fixed c set the reference energy density u_{crit} . Representative $V_0=1$ m³ specifies a calculation volume, not an estimated universe size. Target energy shares t_s on the frozen address moments μ_s star and uniform carrier determine three SI response coefficients η_s .

$$u_{\text{crit}} = \frac{3H_0^2 c^2}{8\pi G}, \eta_s = \frac{u_{\text{crit}} t_s}{\mu_s^*}, [\eta_s] = J m^{-3}. \quad (4)$$

Sector	Arithmetic %	Energy target %	$\eta_s J m^{-3}$
ϕ	5.000000	4.930000	7.5615824991e-10
D	26.800000	26.500000	7.2857613288e-10
R	68.200000	68.570000	7.6461028188e-10

The three η values are explicit calibrations to validated energy targets. Reproduction through calibration and subsequent calculated state responses are recorded separately. An alternate calculation explicitly calibrates to 5%, 26.8%, 68.2% and returns $q_0=-0.523$. It is a changed-target calculation and does not overwrite the inherited reference.

$$\epsilon_s(n, a, \rho) = V_0 \eta_s a^{v_s} g_s(n) L_s, E_s = \sum_n w_n e_s(n) \epsilon_s(n, a, \rho). \quad (5)$$

$$\hat{E}(a) = V_0 \sum_s \eta_s \mu_s a^{v_s} A_s, E = \text{Tr}(\rho \hat{E}) = \sum_s E_s. \quad (6)$$

epsilon_s is the address-response energy before multiplication by address weights and sector effects, in joules. Addresses and the 48 particle slots allocate one energy ledger. Applying the selected 0.8 permutation and normalized generation weights makes the channel-energy sum equal E_phi without multiplying the total by the generation count.

$$V = V_0 a^3, u_s = \frac{E_s}{V}, P_s = - \frac{\partial E_s}{\partial V} \bigg|_{w,e,g,\rho} = - \frac{v_s}{3} u_s. \quad (7)$$

The reference adopts nu_phi=nu_D=0 and nu_R=3. Fixed comoving phenotype and D energies give zero pressure, while R energy proportional to volume gives P_R=-u_R. An independent five-point volume derivative checks the same function. The controls nu_R=0, 1.5, 3 calculate equations of state 0, -0.5, -1. Arithmetic R share alone is not supplied as pressure.

State evolution and expansion from the same operator

At fixed volume, the energy operator divided by E_star=ucrit V0 generates updates in dimensionless clock coordinate xi. This inherits a constitutive clock and does not replace the later shutter-to-proper-time calculation. Noncommuting Kc and Kb transfer energy between D and R, with zero total internal exchange.

$$\begin{aligned} U(\xi) &= e^{-i\xi \hat{E}/E_*}, \rho(\xi) = U \rho_0 U^\dagger, \\ \frac{d\rho}{d\xi} &= -i \left[\hat{E}/E_*, \rho \right], \text{Tr} \left(\hat{E} \frac{d\rho}{d\xi} \right) = 0. \end{aligned} \quad (8)$$

The expansion test integrates the normalized energy generator against log a using the slow constitutive schedule dxi/dt=omega_info, and calculates H from energy density. Pressure is the same volume derivative. At fixed address configuration, cancellation of state exchange gives the continuity relation. The homogeneous flat background and local lens condition retain the scope of the inherited implementation.

Revision r1 incorporates the clock comparison executed in 0.12. At fixed volume xi=E_star tau_phys/hbar gives exactly the same SI energy phase. However, the expansion schedule omega_info=0.7H0 is about 2.10×10⁻⁴³ of E_star/hbar. The old nonuniform histories remain outputs of a slow constitutive schedule and are not identified with physical proper-time evolution of the SI Hamiltonian. Expansion of the uniform I/N state and energy, pressure and local-gravity reproduction are retained.

$$\frac{H}{H_0} = \sqrt{\frac{u}{u_{crit}}}, q = \frac{u+3P}{2u}, \frac{du}{d \log a} + 3(u+P) = 0. \quad (9)$$

$$a_T = c H_0 \sqrt{\frac{u_D}{8u_{crit}}}, \kappa_\theta^2 = \frac{16\pi G}{c^4} (u_D + u_R). \quad (10)$$

a_T enters the calibrated finite local gravity response inherited from 0.5. Lensing is conditional on Phi=Psi in the same patch. The executed joint calculation of twist and expansion from common load remains distinct from a subsequent causal claim that twist produces expansion. At u=0, H=0 and q and energy fractions are null.

Quantity	Inherited reference	Alternate energy target
u J m ⁻³	7.6689477678e-10	7.6689477678e-10
P Pa	-5.2585974844e-10	-5.2302223777e-10
q	-0.528550000	-0.523000000

Quantity	Inherited reference	Alternate energy target
v km s ⁻¹	207.510905127	207.688109769
Deflection arcsec	0.535586511	0.536727827

Address transitions and conversion work ownership

The 0.9 admission order is not yet cosmic evolution in seconds or a physical binary measurement. This stage calculates the required energy exchange for expected S-to-phi transport at fixed carrier state in one representative volume. Pending S receives the same address response as R. Each admitted b_n produces a destination-minus-source energy change and the opposite signed entry in a conversion-work account. This is net exchange between the calculation system and the environment owning conversion work; it is not added as a fourth cosmic fraction. Microscopic environment implementation and physical measurement-apparatus loads remain subsequent transition work.

$$\Delta E_k = \sum_{n \in oddcomp} b_n(k) [\epsilon_\phi(n) - \epsilon_R(n)], \Delta W_k = -\Delta E_k. \quad (11)$$

$$E_{\phi,k} + E_{D,k} + E_{S,k} + E_{R,k} + W_k = constant, \epsilon_S(n) = \epsilon_R(n). \quad (12)$$

The calculated net exchange account at admission completion is 1.3279304601e-12 J. Omitting this exchange makes the original system-energy sum nonconstant and fails the check. Relabeling S as R at recovery has zero exchange because their per-address energies agree and no energy is counted twice. This closes the required exchange quantity and conservation ledger; it does not derive the environmental law driving conversion.

Address rule responses after calibration

When admission count K and phase increment xi change, eta and physical constants remain fixed. The following arithmetic phenotype shares and q values come from the same energy-pressure calculation. They are conditional WRRM_M outputs under changed inputs and are preserved alongside reproduction of the reference universe.

Case	Arithmetic phenotype %	Calculated q
reference	5.000000000	-0.528550000
K4	3.704794619	-0.547908522
K16	5.682473858	-0.518343025
xi_zero	4.859795777	-0.530647725

Verification and reproduction

Checks cover positive energy and direct address sums, three-sector SI calibration, nine inherited cases, alternate energy targets, total-operator positivity, independent five-point pressure differentiation, exponent controls, representative-volume changes, separate address and carrier cutoffs, address-state changes, zero and tiny loads, 48-channel budgets, preserved filter charges, noncommuting updates, homogeneous continuity, four-field expected transport, conversion work and recovery, and invalid-input rejection. After conservation tests pass, a clean copy without saved outputs repeats the calculation.

```
python -m pip install -r calculations/wrra_m_0_10/requirements.txt
python calculations/wrra_m_0_10/run_release.py
```

Parameters.json is the full configuration input and embeds the 0.9 input. Results.json and verification.json record calculation and falsification checks. CSV files contain address energy samples, state-volume results and frame exchange accounts. Run_release.py executes checks and generates both Markdown manuscripts. Word and PDF editions are final publications built from the same sources and are distinguished from raw execution outputs.

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Input SHA256 8342c02c3d872994b13c54c9512d6e7440a45e7ef9602cb1ac4ebe34eddb540f

Subsequent connections and claim scope

The current execution order is 0.10 energy-pressure mapping, 0.11 sequential calibration, 0.12 shutter/proper time/spectra, 0.13 quantization/observation/records, 0.14 cutoffs/capacity/uncertainty, 0.15 stress/curvature/size, 0.16 neutrino mass/mixing/oscillation, 0.17 propagation in the same geometry and 1.0 integration. It preserves the original six-stage goals of quantization, repeated transitions, transition loads and final integration.

The completed result is an executable energy-pressure connection and reproduction of inherited physical outputs under disclosed calibration and constitutive rules. Origins of address responses and energy exponents, general address-carrier entanglement dynamics, a full four-dimensional covariant extension and microscopic measurement apparatus are outside this completed stage. Unmeasured outputs are recorded after model and calibration fixation together with their inputs and falsifiers.

References and public links

Choi Wonsik. WRRM-M 0.9-r1 reviewed 0.7 to 0.9 series. DOI 10.5281/zenodo.23091892.
<https://doi.org/10.5281/zenodo.23091892>

Choi Wonsik. WRRM-M 0.6-r2 Information Load and Twist Gravity and Expansion. DOI 10.5281/zenodo.23076547.
<https://doi.org/10.5281/zenodo.23076547>

WRRM-M repository. <https://github.com/Wonsik-Choi-janefather/wrra-m-0.1>

Minimal Computation Cosmology 2.3.2. <https://github.com/Wonsik-Choi-janefather/minimal-computing-cosmology-2.3.2>